qBraid, MITRE & JonesTrading: GIC 2026

Dynamic Systems Forecasting: Design and benchmark quantum reservoir computing systems to forecast complex, chaotic time-series data for use cases in financial and environmental domains.

Hosted by Connected DMV

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Start
Mar 2026 Wed, ET

Close
Jul 2026 Sat, ET

Phase 1 Deadline

Phase 2 Deadline

qBraid, MITRE & JonesTrading

Launch on qBraid

Quantum Reservoir Computing for Time-Series Intelligence

Dynamic Systems Forecasting

Quantum Reservoir Computing (QRC) is a near-term quantum machine learning paradigm for temporal data processing that requires no gradient-based optimization of the quantum system and is suited to noisy intermediate-scale quantum hardware. Participants will design and benchmark simulation-friendly QRC systems and apply them to real-world industry problems where chaotic dynamics and nonlinear temporal dependencies are central, including financial volatility prediction and climate and weather time-series forecasting. Teams will demonstrate performance across different qubit counts and realistic noise models, benchmark against classical baselines, and implement a common benchmark to validate that the quantum reservoir exhibits sufficient expressivity for forecasting complex, regime-shifting systems.

Click to Download Challenge: Qbraid MITRE JonesTrading_Phase 2 Challenge Description.pdf

Phase 2: Conceptual Design

Phases 2 and 3 of the Global Industry Challenge focus on developing a conceptual design (in Phase 2) to applied execution (in Phase 3), demonstrating technical sophistication and real-world relevance.

Phase 2 focuses on designing and justifying early-stage quantum solutions tailored to each challenge domain. Winning teams are expected to deliver, where possible, functional prototypes or circuit designs, define data preprocessing or encoding workflows, and justify their choice of algorithms and execution platforms, whether simulators or NISQ hardware. Equally important is demonstrating the value of the quantum approach compared to classical alternatives. In addition, teams should provide their estimates of the technical requirements needed to scale their solutions in Phase 3, including the type and extent of quantum hardware and simulator usage required via qBraid's platform, IBM, D-Wave, or QCi. The selection of that device must be made on the Cover Page, or usage will not be granted.

Submission Requirements

Maximum 3 pages PDF (not including references or Cover Page Template), using 11-point Times New Roman font and single spacing, and submit it through the Aqora platform. Only submit one document per team.

Submission must follow GIC requirements: Maximum 3 pages (excluding this cover page and

may not be modified or recreated.

Phase 2 Cover Page Template: [GIC_2026 Cover Page.docx]

Specific requirements are outlined in the Challenge Documents on the Challenge pages.

Generally, your submission should try to address:

- Focus area and rationale
- Technical approach to quantum integration
- Stakeholder relevance
- Data modelling strategy
- Quantum platform justification and resource needs – details for accessing can be found below – selection details coming soon.

Important: Non-compliant submissions due to non-compliant page limit, use of the Cover Page template, or use of AI may be disqualified and voided.

Submission Deadline: Sunday, May 31, 2026, 11:59 PM (EST).

Full Challenge Description:

Quantum Reservoir Computing (QRC) is one of the most promising near-term quantum machine learning paradigms for temporal data processing. Unlike variational quantum algorithms, QRC requires no gradient-based optimization of the quantum system. Only a single linear readout layer is trained, making it naturally suited to today's noisy intermediate-scale quantum hardware. Recent results have demonstrated QRC scaling to 108 qubits on neutral atom hardware with promising results against the classical baselines on chaotic time-series forecasting.

This challenge asks participants to design, build, and benchmark simulation-friendly QRC systems (5–20 qubits) and apply them to real-world industry problems where chaotic dynamics, nonlinear temporal dependencies, and regime shifts make QRC's unique strengths directly relevant. Participants choose one or both of the following application tracks:

1. Financial Volatility Prediction: Markets swing between calm and chaotic periods. Predicting when these shifts happen is a core challenge in quantitative finance, with direct impact on how portfolios are managed and how risk is priced. Build a QRC that detects volatility regime changes in publicly available equity data and forecasts transitions before they happen.
2. Weather Time-Series Forecasting: Atmospheric systems are inherently chaotic, making accurate forecasting one of the classic hard problems in science. Using real-world weather station data from sources like NOAA, build a QRC that predicts temperature, pressure, or humidity sequences. Short-term weather prediction directly impacts agriculture, energy, logistics, and disaster response, making it an ideal testbed for QRC on a problem that matters.
   In addition to the chosen industry track, all participants implement a common MNIST digit classification benchmark using their QRC architecture, providing a standardized comparison across teams and validating that the quantum reservoir exhibits sufficient expressivity.
   Participants are expected to demonstrate how their QRC performs across different qubit counts (e.g., 5, 10, 15 qubits) and under realistic noise models, including depolarizing channels and amplitude damping.

Platform & Compute Access

Participants will receive access to the qBraid platform, which provides a unified environment with quantum simulators, GPU-accelerated training, and quantum processing units (QPUs) from multiple hardware providers. Simulator-based development is the primary mode of work for this challenge; GPU and QPU resources are available to support training and hardware validation runs as participants progress through the challenge.
Alongside their GitHub repository, teams are asked to submit a qBraid Skill: a structured, agent-executable package that allows an AI coding agent to navigate the codebase, configure the reservoir, run training, and reproduce results end-to-end. This is not just documentation; it is a functional interface for reproducibility. Teams are encouraged to leverage qBraid's agentic coding capabilities throughout development, including agent-driven hyperparameter sweeps, automated benchmarking across qubit counts, and AI-assisted noise analysis.

Design Space

QRC offers a rich architectural design space. Participants are free to explore any approach, including but not limited to:
-Hamiltonian design: Ising, Heisenberg, random graphs, Bose-Hubbard, or custom topologies
-Input encoding schemes: amplitude encoding, angle encoding, temporal multiplexing
-Measurement strategies: projective, weak, or continuous measurement; spatial multiplexing across parallel reservoirs
-Feedback mechanisms: re-injecting measurement outcomes to restore fading memory and enhance nonlinearity
-Hybrid architectures: combining classical memory-augmentation layers with quantum feature extraction
-Readout strategies: polynomial regression, ridge regression, or other post-processing to expand feature space

Key References

1. Kornjaca et al., “Large-scale quantum reservoir learning with an analog quantum computer,” arXiv:2407.02553 (2024).
2. Zhu et al., “Practical few-atom quantum reservoir computing,” Phys. Rev. Research 7, 023290 (2025).
3. Tandon et al., "Quantum Reservoir Computing for Corrosion Prediction in Aerospace: A Hybrid Approach for Enhanced Material Degradation Forecasting," arXiv:2505.22837 (2025)
4. Li et al., “Quantum Reservoir Computing for Realized Volatility Forecasting,” arXiv:2505.13933 (2025).
5. Ahmed et al., “Robust quantum reservoir computers for forecasting chaotic dynamics: generalized synchronization and stability,” Proc. R. Soc. A 481 (2025).

Quantum Computer and Simulator Access

To access the qBriad, D-Wave, or QCi platforms in Phase 3, you must fill out the following request form - https://qbraid.typeform.com/to/vTxsKddw

Important Instructions:

1. You must still include your choice and justification in your Phase 2 write-up
2. One Submission Required Per Team for Each Challenge

3. If downselected as a finalist, you will receive access and credits based on the Judging Committee's recommendations.

Submission Deadline: Sunday, May 31, 2026, 11:59 PM (EST).

Questions? Please email quantum@connecteddmv.org

Access Instructions

qBraid [Available for All Challenges]

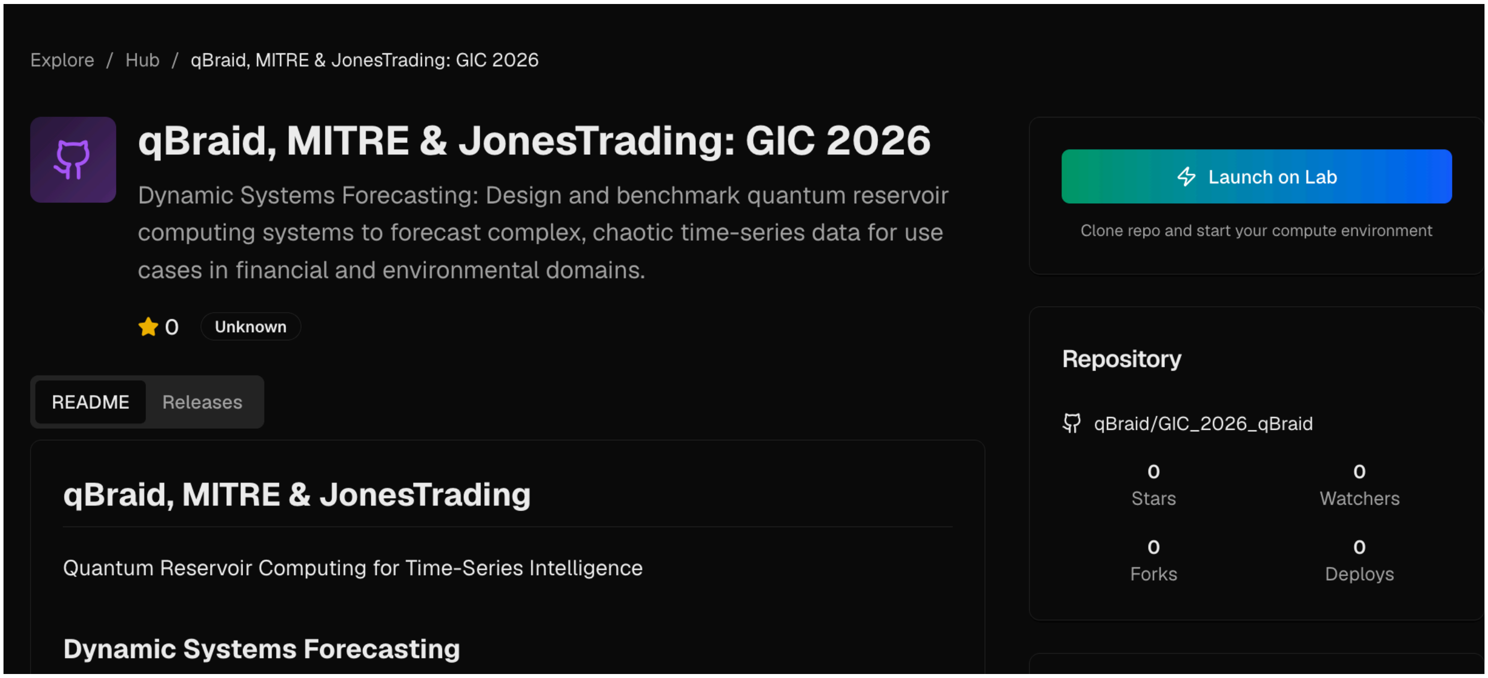
Congrats, challengers, on your journey so far! Now comes the fun part -- running your solutions on real quantum devices.

Challenge finalists (Phase 3) receive access to the following platforms and systems: IonQ AQT, Cepheus-1-108Q, IQM Emerald, IBEX Q, and more. Full list available here once registered: https://account.qbraid.com/devices

If you're using qBraid, all you need to do is click the ‘Launch on qBraid’ button found on Aqora to link your Aqora account with your qBraid account.

Then you should be taken to the challenge page on qBraid (see image). Click Launch on qBraid, and you will have the challenge info cloned into your qBraid account.

As we process device usage requests in the Phase 2 downselect and finalist team selection, we will load your account with credits!



Quantum Computing Inc (QCi) [Available for QCi Challenge Only]

This challenge requires optimization tools which can handle highly nonlinear models. Quantum Computing Inc (QCi) created an analog computing paradigm known as Entropy Quantum Computing (EQC). This paradigm is currently implemented in an electro-optical hybrid system known as Dirac-3. Differentiating it from other analog computing platforms, Dirac-3 offers full connectivity among problem variables with up to fifth degree polynomials. The device also allows the use of two machine-level encodings. One is a discrete encoding- integer valued variables. The other is a quasi-continuous encoding. This flexibility gives the user the ability to solve a wider variety of problems with tighter approximation. While other platforms could be useful for certain aspects of this challenge, Dirac-3 covers the optimization gamut.

Learn about Dirac-3, complete the quick start, then get started with eqc-models to leverage the power of Dirac-3.

QCi's Dirac-3 - https://quantumcomputinginc.com/products/commercial-products/dirac-3
Dirac Quick Start - https://quantumcomputinginc.com/learn/developer-resources/entropy-quantum-optimization/dirac-quick-...
eqc-models - https://pypi.org/project/eqc-models/

eqc-models-tutorial - https://github.com/qci-wdyk/eqc-models-tutorial
QCi Resource Library - https://quantumcomputinginc.com/learn

D-Wave [Available for All Challenges]

Learning about our platform:

D-Wave offers a variety of free and paid trainings, e-books, and webinars. A few of the most relevant ones are listed below:
• Quantum Fit webinar: https://youtu.be/8ewcmvYDRV8?si=mQGiBpK8jjY7Vd6S
• Quantum Fit panel from Qubits 2026: https://youtu.be/E4aJGmsNvog?si=k9MNIH13RpM5YtRL
• Quantum Fit eBook - this is an asset that we offer for Commercial organizations, but could be used by your applicants too, to assess if their project would benefit from using our platform: Is Quantum Optimization a Fit for Your Business?
• Free Introduction to Quantum Computing course (if any participants need to understand annealing vs gate and use cases for each): https://training.dwavequantum.com/product?catalog=IntroductionQuantumComputing
• Getting Started with Applied Quantum Optimization course covers problem discovery and problem formulation, this is a paid training, the price is $50.
https://training.dwavequantum.com/product?catalog=GettingStartedAppliedQuantumOptimization
• Additional, more advanced trainings can be found here: D-Wave | Courses

Accessing the platform:

• GIC organizers will provide D-Wave access and provide the users of the finalist teams access to the platform.
• Users will be able to access either QPU or D-Wave's Hybrid Solvers
• There will be a set time limit and credits determined by the organizing and judging committee.
• There is an important limitation on access; the users need to be in a so-called "Leap Approved" country. Please find the list of these countries here: From What Countries Can I Access D-Wave's Leap Quantum Cloud Service? – D-Wave Quantum Inc
• Please indicate you meet these needs in the Typeform.

IBM Quantum Open Plan [Available for All Challenges]

As highlighted in IBM's recent updates, the Open Plan has been significantly expanded and is intentionally designed to support early stage exploration, prototyping, and algorithm work.

In particular:
• The Open Plan is free and publicly available.
• Eligible users can unlock expanded runtime (up to 180 minutes over 12 months) once a modest usage threshold is reached, allowing teams to scale experimentation as their work matures.
• This expansion is structured so that additional access becomes available at an appropriate usage trigger point, without special forms, approvals, or event specific onboarding.

IBM believes this model better aligns with the likely needs and maturity of participating teams.

For participants seeking details on platform capabilities and access, these should be the primary references:
• https://www.ibm.com/quantum/blog/open-plan-updates
• https://www.ibm.com/quantum/blog/whats-new-q1-2026

Any questions about accessing or using IBM Quantum Open Plan should be directed to IBM.